Supplementary Material: Urban Settlement Indices Catalogue and Implementation Details

This supplement provides the full 46 index list with formulas, band mapping, required preprocessing, the canonical thresholding rule reported in the source study, and the original citations.

A. *Summary of Urban Settlement Indices*

Urban settlement research now relies on an extensive toolbox of 46 indices that span multispectral, panchromatic, thermal, night time light (NTL) and hybrid formulations.To show methodological breadth, one representative indice from each family is then profiled and its design rationale and trade offs are discussed.

*1) **Normalized Difference Built-Up Indice (NDBI)**:* NDBI[1] leverages a characteristic contrast of built-up surfaces, with higher SWIR reflectance that reflects low moisture and roughness and comparatively lower NIR than healthy vegetation. This contrast is used to delineate impervious areas with only two widely available bands, such as Landsat and Sentinel-2. Operationally, the indice is computed per pixel and thresholded, with pixels marked as built-up when $\mathrm{NDBI} > \tau$. The threshold τ can be a fixed value or estimated with unsupervised methods to adapt across scenes. Strengths include simplicity, broad sensor compatibility and fast large area processing. Common failure modes include confusion with bright bare soils or salt flats and illumination or seasonal effects that shift SWIR–NIR contrast. For this reason NDBI is often paired with vegetation or water suppressors or embedded in composite indices such as IBI[2].

$$\mathrm{NDBI} = \frac{\rho_{\mathrm{SWIR}_1} - \rho_{\mathrm{NIR}}}{\rho_{\mathrm{SWIR}_1} + \rho_{\mathrm{NIR}}} \tag{1}$$

*2) **VrNIR-BI (Visible-Red–NIR Built-Up Indice)**:* VrNIR-BI[3] targets the dry vegetation confusion that is common to NDBI and IBI[2] by exploiting the strong Red–NIR contrast of many impervious materials. These materials often have higher Red reflectance than healthy vegetation and comparatively lower NIR. In Landsat-7 and Landsat-8 tests, VrNIR-BI delivered overall accuracy above 92% with an 8–10 pp Kappa gain over classical indices in semi arid and high density urban settings[4], [3]. Limitations include sensitivity of the Red band to haze, aerosols and illumination, possible confusion with bright bare soils or oxidised surfaces and shadow induced NIR suppression. Scene specific thresholding, for example Otsu or k-means, and pairing with vegetation or water suppressors where available can reduce these errors.

$$\mathrm{VrNIR\text{-}BI} = \frac{\rho_{\mathrm{Red}} - \rho_{\mathrm{NIR}}}{\rho_{\mathrm{Red}} + \rho_{\mathrm{NIR}}} \tag{2}$$

*3) **VgNIR-BI (Visible-Green–NIR Built-Up Indice)**:* VgNIR-BI[3] substitutes Green for the Red band in classic Red–NIR ratios. It exploits the relatively high Green and low NIR signature of many impervious materials to separate built-up land from dry vegetation, particularly in semi arid settings and on sensors that lack SWIR. In Landsat-7 and Landsat-8 tests, VgNIR-BI achieved overall accuracies above 92% with 8–10 pp Kappa gains over NDVI based baselines[3]. Reliance on the Green band makes it sensitive to haze and illumination. Bright bare soils, rock and painted or metallic rooftops can mimic the target contrast, and deep shadows may suppress the NIR response. In practice, scene specific thresholding and, where possible, pairing with vegetation or water suppressors are recommended to reduce false positives and false negatives.

$$\mathrm{VgNIR\text{-}BI} = \frac{\rho_{\mathrm{Green}} - \rho_{\mathrm{NIR}}}{\rho_{\mathrm{Green}} + \rho_{\mathrm{NIR}}} \tag{3}$$

*4) **BBI (Built-Up and Bare-Land Areas Indice)**:* BBI[5] addresses the common confusion between impervious surfaces and bare soil by thresholding two chromatic ratios derived from NDBI, a Blue–Green ratio and a Red–Green ratio, and then summing the resulting binary masks. The idea is that built-up and bare land pixels tend to be relatively bright in Blue or Red compared to Green. Concurrent exceedance flags likely impervious or bare materials while vegetation is suppressed. On Landsat-8 OLI test sites, BBI reported accuracy gains above 30 pp over classical NDBI in semi arid settings[5]. Limitations include sensitivity of the Blue channel to haze and atmospheric scattering, illumination and shadow effects that depress Red or Blue responses and potential false positives from bright sands, salt pans or painted and metal rooftops. Per scene thresholds T also govern transferability. Pairing with vegetation or water suppressors and using data driven threshold selection is therefore recommended.

$$\mathrm{NDBI}_{\text{Blue-Green}} = \frac{B - G}{B + G}, \tag{4}$$

$$\mathrm{NDBI}_{\text{Red-Green}} = \frac{R - G}{R + G}, \tag{5}$$

$$\mathrm{NDBI}_b = \begin{cases} 1 & \text{if NDBI} > T, \\ 0 & \text{otherwise,} \end{cases} \tag{6}$$

$$\mathrm{BBI} = \mathrm{NDBI}_{\text{Blue-Green},b} + \mathrm{NDBI}_{\text{Red-Green},b} \tag{7}$$

*5) **CBCI (Combinational Biophysical Composition Indice)**:* CBCI[6] integrates a Modified Built-up Soil Indice (MBSI) with the Optimized Soil Adjusted Vegetation Indice (OSAVI). The two terms are balanced by an empirical factor $A = 0.51$ and used to separate impervious, vegetation, soil and water using only Green, Red and NIR bands. This design is well suited to high resolution optical sensors such as Sentinel-2 MSI and GF-1[7]. Intuitively, MBSI enhances brightness contrasts between built-up areas and soil, while OSAVI suppresses vegetation. Their weighted combination improves class separation, with reported correlations $r > 0.84$ to impervious surface area across multiple sensors[6]. Limitations include sensitivity of the Red and Green channels to haze and illumination, phenology dependent variability introduced by the OSAVI term, potential confusion with bright bare soils, sands or painted and metal roofs and dependence on scene specific thresholding for binary masks. In practice, recalibrating A if needed, using data driven thresholds such as Otsu or k-means and optionally pairing with water or vegetation suppressors can reduce errors.

$$\mathrm{CBCI} = (1 + A)\,\mathrm{MBSI} - A\,\mathrm{OSAVI}, \tag{8}$$

$$\mathrm{MBSI} = \frac{2\,(R - G)}{(R + G) - 2}, \tag{9}$$

$$\mathrm{OSAVI} = \frac{NIR - R}{NIR + R + 0.16}, \tag{10}$$

where R, G and NIR are red, green and near infrared reflectance, respectively.

*6) **WE-NDBI (Water-Eraser Normalized Difference Built-Up Indice):*** WE-NDBI[8] adapts classical NDBI for water rich deltaic and peri urban settings. A Red–Green chromatic ratio is used as a water mask and high water like responses are set to zero, which reduces false positives along rivers, canals and wetlands[8]. On GF-1 WFV imagery, the method reported 91.4% extraction accuracy, about 11 pp higher than BBI-OLI under comparable conditions[8]. Limitations include sensitivity of the Red and Green bands to haze and illumination, dependence on the scene specific water threshold W, which affects transferability, and potential confusion in turbid or sediment laden waters as well as with bright roofs or sand. Estimating W with data driven methods and, where available, pairing with a dedicated water suppressor can improve robustness.

$$\mathrm{NDBI}_{RG} = \frac{R - G}{R + G}, \tag{11}$$

$$\text{WE-NDBI} = \begin{cases} \mathrm{NDBI}_{RG}, & \mathrm{NDBI}_{RG} \le W, \\ 0, & \mathrm{NDBI}_{RG} > W, \end{cases} \tag{12}$$

with W an empirically tuned water threshold.

*7) **PISI (Perpendicular Impervious Surface Indice):*** PISI[9] rotates the Blue–NIR spectral space to a discriminant axis that maximises contrast between impervious surfaces and vegetation or soil. This approach is similar to projecting onto a line that is perpendicular to the vegetation response. Applied to Landsat-8 scenes in Wuhan, Shenyang, Xining and Guangzhou, PISI achieved 89–97% overall accuracy and outperformed NDBI and BCI[10], while requiring only two bands. This makes it suitable for sensors that lack SWIR or TIR. Limitations include sensitivity of the Blue band to haze, aerosols and illumination, potential confusion with bright bare soils or painted and metal roofs and dependence on scene and sensor specific coefficients and thresholds that may need recalibration. In practice, adaptive thresholding, for example Otsu or k-means, and optional vegetation or water suppressors can improve robustness.

$$\mathrm{PISI} = 0.8192\,\rho_{\mathrm{Blue}} - 0.5735\,\rho_{\mathrm{NIR}} + 0.0750 \tag{13}$$

*8) **UI (Urbanization Indice):*** The Urbanization Indice (UI)[11] estimates built-up intensity from the contrast between short wave infrared and near infrared reflectance. Impervious materials typically show $\mathrm{SWIR}_2 > \mathrm{NIR}$. UI expresses this contrast as a scaled ratio and was originally devised for Landsat TM and ETM+. In validation, for example in Colombo, Sri Lanka, UI correlates strongly with proxies of urban density such as population, land value and energy consumption. Reported empirical relationships include $UI = 0.47\,UD + 49.2$ and $UD = 1.65\,UI - 72.74$[11]. Strengths include simplicity, broad sensor compatibility and a monotonic response to urban density. Limitations include sensitivity to shadows and illumination and to seasonal effects, dependence on spatial resolution and sensor band definitions, which requires scene specific thresholds, and confusion with bright bare soils or dry substrates in arid regions.

$$\mathrm{UI} = \left(\frac{\rho_{\mathrm{SWIR}_2} - \rho_{\mathrm{NIR}}}{\rho_{\mathrm{SWIR}_2} + \rho_{\mathrm{NIR}}} + 1 \right) \times 100 \tag{14}$$

*9) **NBI (New Built-up Indice):*** NBI[12] combines Red and SWIR_1 reflectance and normalises by NIR. This suppresses vegetation response and at the same time emphasises chromatic (Red) and moisture or roughness (SWIR) contrasts that are typical of many impervious materials. In Landsat TM experiments it reported 70.27% overall accuracy with a Kappa of 0.3965[12]. Strengths include simplicity, reliance on three widely available bands and partial mitigation of vegetation driven false positives. Common limitations are sensitivity to illumination, soil brightness and moisture variability, possible confusion with bright sands, salt flats or painted and metal roofs, and dependence on scene specific thresholding. Performance can decline in semi arid settings where soil spectra overlap with built-up signatures.

$$\mathrm{NBI} = \frac{\rho_{\mathrm{Red}} \cdot \rho_{\mathrm{SWIR}_1}}{\rho_{\mathrm{NIR}}} \tag{15}$$

*10) **NBAI (Normalized Built-Up Area Indice):*** The NBAI[13] augments built-up detection by contrasting a ratio of SWIR_1 to Green reflectance, which suppresses vegetation and moderates bright soil effects, against SWIR_2 in a normalized difference. This enhances impervious signals in semi arid or transitional zones. Applied to Landsat TM and ETM+ scenes over Islamabad, NBAI achieved 81.18% overall accuracy with a Kappa of 0.63, about 10 to 13% higher than NDBI and NBI[13]. Strengths include reliance on widely available bands (SWIR_1, SWIR_2, Green) and improved separation where soil brightness confuses simpler indices. Limitations include sensitivity of the Green band to haze and illumination, instability of the SWIR_1/Green ratio in deep shadow or very low Green backgrounds, potential confusion with bright sands, salt flats and some painted or metal roofs, and dependence on scene specific thresholding. In practice, a water mask to avoid bright water misclassification, optional vegetation or water suppressors and data driven threshold selection, for example Otsu or k-means, improve robustness.

$$\mathrm{NBAI} = \frac{\rho_{\mathrm{SWIR}_2} - \frac{\rho_{\mathrm{SWIR}_1}}{\rho_{\mathrm{Green}}}}{\rho_{\mathrm{SWIR}_2} + \frac{\rho_{\mathrm{SWIR}_1}}{\rho_{\mathrm{Green}}}} \tag{16}$$

*11) **BRBA (Band Ratio for Built-up Area)**:* BRBA[13] uses a ratio of Red to SWIR_1 to emphasise impervious materials that tend to be relatively bright in Red and darker in SWIR_1. This gives a simple and sensor compatible discriminator on Landsat and Sentinel imagery. Applied over Islamabad, BRBA achieved 90.13% overall accuracy and outperformed NDBI (84.14%) and NBI (70.27%) under comparable conditions[13]. Limitations include sensitivity to illumination and haze that affect Red, confusion with bright bare soils, sands or painted and metal roofs and dependence on scene specific thresholding. Using a water mask to avoid bright water artefacts and adopting data driven thresholds, for example Otsu or k-means, improves robustness.

$$\mathrm{BRBA} = \frac{\rho_{\mathrm{Red}}}{\rho_{\mathrm{SWIR}_1}} \tag{17}$$

*12) **BAEI (Built-up Area Extraction Indice)**:* BAEI[14] enhances built-up detection with a stabilised chromatic ratio that emphasises high Red over lower Green and SWIR_1 reflectance. The constant $L = 0.3$ improves numerical stability and moderates illumination effects. The indice is computationally light and compatible with Landsat 8 and Sentinel-2. Tested in Djelfa, Algeria, BAEI achieved 92.66% overall accuracy with a Kappa of 0.8537[14]. Applying a vegetation mask further improves reliability by reducing spectral mixing[14]. Limitations include sensitivity of the Green band to haze and aerosols, possible confusion with bright bare soils, sands or painted and metal roofs and dependence on scene specific thresholding. Data driven thresholds, for example Otsu or k-means, and optional water or vegetation suppressors help to reduce these issues.

$$\mathrm{BAEI} = \frac{\rho_{\mathrm{Red}} + L}{\rho_{\mathrm{Green}} + \rho_{\mathrm{SWIR}_1}}, \quad L = 0.3 \tag{18}$$

*13) **BUI (Built-Up Indice)**:* BUI[15] detects impervious surfaces using two normalised contrasts, Red versus SWIR_1 and SWIR_2 versus SWIR_1. In this way it emphasises built-up materials without relying on NIR, which is more sensitive to shadows and phenology. This design improves separability in mixed urban and rural mosaics and under cast shadows. In case studies over Thessaloniki and Patras in Greece it achieved around 90% overall accuracy[15]. Limitations include potential confusion with bright bare soils and construction sites, dependence on scene specific thresholding, sensitivity to SWIR calibration and atmospheric conditions such as haze and moisture and reduced performance over highly reflective roofs or salt and sand surfaces. Pairing with vegetation or water suppressors and data driven thresholds, for example Otsu or k-means, is recommended.

$$\mathrm{BUI} = \left(\frac{\rho_{\mathrm{Red}} - \rho_{\mathrm{SWIR}_1}}{\rho_{\mathrm{Red}} + \rho_{\mathrm{SWIR}_1}}\right) + \left(\frac{\rho_{\mathrm{SWIR}_2} - \rho_{\mathrm{SWIR}_1}}{\rho_{\mathrm{SWIR}_2} + \rho_{\mathrm{SWIR}_1}}\right) \tag{19}$$

*14) **SwiRed (Shortwave Infrared and Red Ratio Indice)**:* SwiRed[16] identifies built-up areas using a SWIR_1 and Red normalised contrast that does not rely on NIR. This makes it suitable for multi temporal mapping on platforms such as Landsat 5 to 8 and Sentinel-2. The rationale is that many impervious materials show relatively higher SWIR_1 than Red compared with vegetation, which supports a simple chromatic discriminator. In practice, empirical thresholds such as $0 < \mathrm{SwiRed} < 0.22$ have been used to separate classes[16]. Related approaches that address the same lack of NIR include BRBA[13] and BUI[15]. Limitations include sensitivity of the Red band to haze and illumination, possible confusion with bright bare soils, sands or painted and metal roofs and dependence on scene specific thresholding. Applying vegetation and water masks and using data driven thresholds, for example Otsu or k-means, can improve robustness.

$$\mathrm{SwiRed} = \frac{\rho_{\mathrm{SWIR}_1} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{SWIR}_1} + \rho_{\mathrm{Red}}} \tag{20}$$

*15) **ENDISI (Enhanced Normalized Difference Impervious Surfaces Indice)**:* ENDISI[17] combines a Blue reflectance term with a $\mathrm{SWIR}_1/\mathrm{SWIR}_2$ ratio and a squared MNDWI component inside a scaled normalised difference. The scene adaptive coefficient α, which is estimated from band means, balances these parts. Bright soils are down weighted by the SWIR ratio and open water is suppressed by $(\mathrm{MNDWI})^2$. This improves discrimination among impervious surfaces, soil, vegetation and water using standard optical bands. In evaluation tests, ENDISI achieved 93.9% overall accuracy with SDI equal to 1.499[17]. Limitations include sensitivity of the Blue band to haze and atmospheric effects, dependence on accurate reflectance calibration and scene specific α and thresholds, which can affect transferability, potential confusion over vegetated artificial surfaces such as green roofs or turf or very bright dry soils and salt flats and the requirement for both SWIR bands, which limits use on sensors without SWIR_2. Data driven thresholds, for example Otsu or k-means, optional water or vegetation masks and light morphological filtering can improve robustness.

$$\mathrm{ENDISI} = \frac{\rho_{\mathrm{Blue}} - \alpha \times \left[\frac{\rho_{\mathrm{SWIR}_1}}{\rho_{\mathrm{SWIR}_2}} + (\mathrm{MNDWI})^2\right]}{\rho_{\mathrm{Blue}} + \alpha \times \left[\frac{\rho_{\mathrm{SWIR}_1}}{\rho_{\mathrm{SWIR}_2}} + (\mathrm{MNDWI})^2\right]} \tag{21}$$

$$\alpha = \frac{2 \times (\rho_{\mathrm{Blue}})_{\mathrm{Mean}}}{\left[\left(\frac{\rho_{\mathrm{SWIR}_1}}{\rho_{\mathrm{SWIR}_2}}\right)_{\mathrm{Mean}} + (\mathrm{MNDWI}^2)_{\mathrm{Mean}}\right]} \tag{22}$$

$$\mathrm{MNDWI} = \frac{\rho_{\mathrm{Green}} - \rho_{\mathrm{SWIR}_1}}{\rho_{\mathrm{Green}} + \rho_{\mathrm{SWIR}_1}} \tag{23}$$

*16) $\mathbf{BU}_b$ **(Binary Built-Up Indice)**:* $\mathbf{BU}_b$[18] creates a binary impervious surface mask by thresholding NDVI, which acts as a vegetation suppressor, and NDBI, which acts as a built-up enhancer, and then differencing the two masks. This emphasises pixels with high NDBI and low or negative NDVI. In the original formulation the binary masks are written to 8 bit values (0 or 254). This supports fast large area processing and efficient multi temporal change detection on Landsat ETM+. Strengths include simplicity and low computational cost. Limitations arise from the rigid zero thresholds and hard binarisation, which can misclassify spectrally ambiguous targets such as bright bare soils, salt or sand and painted or metal roofs and areas affected by shadows, haze or seasonal shifts that flip NDVI or NDBI signs. Replacing fixed thresholds with data driven ones, for example Otsu or k-means, applying water and vegetation masks and using light morphological filtering can improve robustness.

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}}}, \tag{24}$$

$$\mathrm{NDVI}_b = \begin{cases} 254 & \text{if NDVI} > 0, \\ 0 & \text{otherwise,} \end{cases} \tag{25}$$

$$\mathrm{NDBI}_b = \begin{cases} 254 & \text{if NDBI} > 0, \\ 0 & \text{otherwise,} \end{cases} \tag{26}$$

$$\mathbf{BU}_b = \mathrm{NDBI}_b - \mathrm{NDVI}_b \tag{27}$$

17) $\mathbf{BU}_c$ ***(Built-up Continuous Indice):*** $\mathbf{BU}_c$[18] combines continuous NDBI, which acts as a built-up enhancer, and NDVI, which acts as a vegetation suppressor, without hard thresholding. This gives a graded response that sharpens contrast between urban and non urban areas, especially in heterogeneous mosaics where binary masks fragment. Reported evaluations show gains of up to 20% over NDBI in overall accuracy[18]. Strengths include simplicity, preservation of sub pixel variation and compatibility with standard Landsat and Sentinel bands. Limitations arise from sensitivity to illumination and atmospheric conditions, radiometric calibration differences between scenes and sensors and residual confusion with bright bare soils or sand. Light radiometric normalisation, data driven thresholding if a binary mask is needed and optional vegetation and water masks can improve robustness.

$$\mathbf{BU}_c = \left(\frac{\rho_{\mathrm{SWIR}_1} - \rho_{\mathrm{NIR}}}{\rho_{\mathrm{SWIR}_1} + \rho_{\mathrm{NIR}}}\right) - \left(\frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}}}\right) \tag{28}$$

*18)**IBI (Index-based Built-up Indice):*** IBI[2] integrates a built up enhancer (NDBI) with vegetation and water suppressors (NDVI or SAVI, and MNDWI) inside a normalised difference form. This design reduces false positives near vegetation and water in mixed urban mosaics. The NDVI based version is common where canopy cover is substantial, while the SAVI variant with $L = 0.5$ improves stability in sparsely vegetated or arid scenes. Computed per pixel and then thresholded so that IBI $> \tau$ marks built up land, IBI uses only visible, NIR and SWIR bands that are available on Landsat and Sentinel. Strengths include better suppression of vegetated and water pixels and wide sensor compatibility. Limitations include dependence on accurate atmospheric correction and cloud or shadow masks, the quality of the MNDWI term in turbid or shallow waters, the need for SWIR_1 and Green bands for MNDWI, and scene specific thresholding that affects transferability. In practice, data driven choices of τ, for example Otsu or k-means, and optional water or vegetation masks further improve robustness.

$$\mathrm{IBI}_{\mathrm{NDVI}} = \frac{2 \cdot [\mathrm{NDBI} - (\mathrm{NDVI} + \mathrm{MNDWI})]}{2 \cdot [\mathrm{NDBI} + (\mathrm{NDVI} + \mathrm{MNDWI})]} \tag{29}$$

$$\mathrm{IBI}_{\mathrm{SAVI}} = \frac{2 \cdot [\mathrm{NDBI} - (\mathrm{SAVI} + \mathrm{MNDWI})]}{2 \cdot [\mathrm{NDBI} + (\mathrm{SAVI} + \mathrm{MNDWI})]} \tag{30}$$

$$\mathrm{SAVI} = \frac{(1 + L) \cdot (\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}})}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}} + L}, \quad L = 0.5 \tag{31}$$

*19)**VIBI (Vegetation Index Built-up Indice):*** VIBI[20] combines NDVI, which captures vegetation response, and NDBI, which captures impervious response, into a single ratio. This gives a simple sensor compatible discriminator for vegetated urban mosaics without ancillary data or supervised training. VIBI tends toward 1 where vegetation dominates and moves toward 0 where built up signals prevail. Applied to Landsat imagery over Volos and Chania in Greece, reported accuracies were 91.1% and 91.9% with Kappa values of 0.56 and 0.51. Post classification smoothing increased Kappa to 0.66[20]. The ratio structure brings some limitations. The indice is undefined when $\mathrm{NDVI} + \mathrm{NDBI} = 0$ and becomes numerically unstable when $|\mathrm{NDVI} + \mathrm{NDBI}|$ is small. Performance is sensitive to atmospheric and illumination conditions, shadows and mixed pixels. VIBI also requires the bands needed to compute both NDVI (Red, NIR) and NDBI (NIR, SWIR). In practice, guarding the denominator with a small ε, applying water or vegetation masks and using data driven thresholds when a binary map is needed can improve robustness.

$$\mathrm{VIBI} = \frac{\mathrm{NDVI}}{\mathrm{NDVI} + \mathrm{NDBI}} \tag{32}$$

20) ***BCI (Biophysical Composition Indice):*** BCI[10] maps mixed urban pixels by combining Tasseled Cap Transformation (TCT) components, brightness (H), greenness (V) and wetness (L)[21], in a normalised contrast. Impervious materials tend to be bright and relatively dry with higher H and L, while vegetation is characterised by high V. The ratio therefore yields larger values for built up areas and negative values for vegetation. Using sensor specific TCT coefficients and min to max normalisation, BCI can be applied to Sentinel-2 MSI with the updated parameters of Shi and Xu (2021)[19]. The required Sentinel-2 coefficients are listed in Supplementary Table **S**9. Strengths include improved separation in complex mixed pixels where NDVI or NDBI can fail. Limitations include dependence on accurate atmospheric correction and TCT coefficients, sensitivity of L to recent rainfall or soil moisture that can cause confusion with wet soils or shallow water, numerical instability when the denominator is small and the need for scene specific thresholds when a binary mask is required. Guarding the denominator with a small ε is recommended.

$$\mathrm{BCI} = \frac{(H + L)/2 - V}{(H + L)/2 + V} \tag{33}$$

$$H = \frac{\mathrm{TC1} - \mathrm{TC1}_{\mathrm{min}}}{\mathrm{TC1}_{\mathrm{max}} - \mathrm{TC1}_{\mathrm{min}}} \tag{34}$$

$$V = \frac{\mathrm{TC2} - \mathrm{TC2}_{\mathrm{min}}}{\mathrm{TC2}_{\mathrm{max}} - \mathrm{TC2}_{\mathrm{min}}} \tag{35}$$

$$L = \frac{\mathrm{TC3} - \mathrm{TC3}_{\mathrm{min}}}{\mathrm{TC3}_{\mathrm{max}} - \mathrm{TC3}_{\mathrm{min}}} \tag{36}$$

*21)**RNDSI (Ratio Normalized Difference Soil Indice):*** RNDSI[22] aims to isolate bare soil in urban mosaics by combining a soil sensitive normalised difference (NDSI, SWIR_2 compared with Green) with Tasseled Cap brightness (TC1). First, NDSI and TC1 are min to max normalised to NNDSI and NTC1. RNDSI is then formed as NNDSI/NTC1, which emphasises pixels with strong soil

Supplementary Table S9: Sentinel-2 MSI tasseled cap transformation (TCT) coefficients for the 12 band configuration (all bands except the cirrus band, B10). Rows give the linear combination weights used to compute the Brightness (B), Greenness (G) and Wetness (W) components from at sensor reflectance (Level-1C). Columns list the contributing MSI bands (coastal aerosol through SWIR2). These coefficients are commonly used to compress multispectral data and emphasise urban or bare soil (B), vegetation vigor (G) and surface moisture or structure (W). Coefficients adapted from [19].

Type	TCT	Coastal	Blue	Green	Red	RE-1	RE-2	RE-3	$\mathbf{NIR_1}$	$\mathbf{NIR_2}$	WV	$\mathbf{SWIR_1}$	$\mathbf{SWIR_2}$
	B	0.2381	0.2569	0.2934	0.3020	0.3099	0.3740	0.4180	0.3580	0.3834	0.0103	0.0896	0.0780
12 Bands	**G**	-0.2266	-0.2818	-0.3020	-0.4283	-0.2959	0.1602	0.3127	0.3138	0.4261	0.1454	-0.1341	-0.2538
	W	0.1825	0.1763	0.1615	0.0486	0.0170	0.0223	0.0219	-0.0755	-0.0910	-0.1369	-0.7701	-0.5293

B: Brightness, **G**: Greenness, **W**: Wetness, **RE**: Red Edge, **NIR**: Near Infrared, **WV**: Water Vapor, **SWIR**: Short Wave Infrared.

contrast and moderates responses that are dominated only by brightness, for example highly reflective impervious or salt surfaces. This requires sensors that provide SWIR_2, Green and valid TCT coefficients, such as Landsat or Sentinel-2 with sensor specific TCT. Reported applications show utility in heterogeneous terrain[22]. Limitations include sensitivity to radiometric calibration and to the scene wise min to max choices, numerical instability where $\text{NTC1} \approx 0$ and confusion in very bright sands or newly constructed materials. Data driven thresholding, robust percentile normalisation and optional water or vegetation masks can improve robustness.

$$\text{NDSI} = \frac{\rho_{\text{SWIR}_2} - \rho_{\text{Green}}}{\rho_{\text{SWIR}_2} + \rho_{\text{Green}}} \tag{37}$$

$$\text{NNDSI} = \frac{\text{NDSI} - \min(\text{NDSI})}{\max(\text{NDSI}) - \min(\text{NDSI})} \tag{38}$$

$$\text{NTC1} = \frac{\text{TC1} - \min(\text{TC1})}{\max(\text{TC1}) - \min(\text{TC1})} \tag{39}$$

$$\text{RNDSI} = \frac{\text{NNDSI}}{\text{NTC1}} \tag{40}$$

22) ***CBI (Combinational Built-Up Indice):*** CBI[23] integrates three normalised cues to enhance impervious surface detection. These are PC1, the first principal component from Red, Green and NIR as a proxy for high albedo built materials, NDWI, which suppresses water and shadows, and SAVI, which reduces vegetation influence. After min to max scaling to comparable ranges, CBI forms a normalised contrast that boosts $\text{PC1}_{\text{norm}} + \text{NDWI}_{\text{norm}}$ and down weights $\text{SAVI}_{\text{norm}}$. This improves separation in mixed urban mosaics on Landsat and Sentinel imagery. Strengths include reliance on widely available optical bands and explicit suppression of vegetation and water. Limitations include sensitivity to PC1 estimation, outlier effects in min to max normalisation, illumination and haze impacts on the visible bands and possible denominator instability. Using percentile based scaling, adding a small ε to the denominator when needed and adopting data driven thresholds can improve transferability.

$$\text{NDWI} = \frac{\rho_{\text{Green}} - \rho_{\text{NIR}}}{\rho_{\text{Green}} + \rho_{\text{NIR}}} \tag{41}$$

$$\text{CBI} = \frac{(\text{PC1}_{\text{norm}} + \text{NDWI}_{\text{norm}}) - 2 \cdot \text{SAVI}_{\text{norm}}}{(\text{PC1}_{\text{norm}} + \text{NDWI}_{\text{norm}}) + 2 \cdot \text{SAVI}_{\text{norm}}} \tag{42}$$

23) ***BLFEI (Built-up Land Features Extraction Indice):*** BLFEI[24] contrasts SWIR_1 against the mean of Green, Red and SWIR_2. This strengthens separation of impervious materials from vegetation and barren land on medium resolution sensors such as Landsat-8 OLI3, OLI4, OLI6 and OLI7. The idea is that many built surfaces show relatively higher SWIR_1 reflectance and distinct SWIR_1 to SWIR_2 behaviour compared with vegetation. Histogram analyses report strong class separability, for example SDI between built and vegetation equal to 3.25 under varying illumination[24]. Strengths include simplicity, use of widely available optical and SWIR bands and robustness across typical daylight conditions. Limitations include sensitivity to atmospheric haze that affects visible bands, possible confusion with bright dry soils, salt pans or painted and metal roofs, dependence on scene specific thresholding, the requirement for both SWIR bands, which limits use on sensors without SWIR_2, and the need for water masking in deltaic or coastal scenes. Light radiometric normalisation, data driven thresholds such as Otsu or k-means and optional vegetation or water masks can improve transferability.

$$\text{BLFEI} = \frac{\left(\frac{\rho_{\text{Green}} + \rho_{\text{Red}} + \rho_{\text{SWIR}_2}}{3}\right) - \rho_{\text{SWIR}_1}}{\left(\frac{\rho_{\text{Green}} + \rho_{\text{Red}} + \rho_{\text{SWIR}_2}}{3}\right) + \rho_{\text{SWIR}_1}} \tag{43}$$

24) ***MBAI (Modified Built-Up Area Indice):*** MBAI[25] enhances built-up detection in arid and semi arid settings by combining NIR with weighted Green and SWIR_1 reflectance in a stabilised ratio. The numerator emphasises chromatic Green and moisture or roughness SWIR_1 contrasts that are typical of impervious materials, while the $1 + \text{NIR}$ denominator dampens vegetation response and moderates illumination effects. Originally calibrated on SPOT imagery over Algerian sites, MBAI reduced confusion with bare soil that is found in simpler indices such as NDBI and IBI and reported overall accuracies of 95% for Laghouat and 91% for M'Sila[25]. Strengths include good performance in dry regions and a continuous output that can be rank ordered without a fixed threshold. Limitations include sensor specific coefficients, so recalibration is advised for Landsat and Sentinel, sensitivity to atmospheric and illumination conditions, potential confusion over very bright sands or salt flats or painted and metal roofs and reliance on SWIR_1, which restricts use on sensors that lack that band. Optional water and vegetation masks and data driven thresholds, when binarisation is desired, can improve robustness.

$$\text{MBAI} = \frac{\rho_{\text{NIR}} + 1.57 \times \rho_{\text{Green}} + 2.4 \times \rho_{\text{SWIR}_1}}{1 + \rho_{\text{NIR}}} \tag{44}$$

*25) **UBI (Urban Built-Up Indice)**:* UBI[26] forms a synthetic RGB composite from NIR, SWIR and a scaled Red channel and then converts it to HSV space. The indice is a normalised hue to value contrast, $(H-V)/(H+V)$, that emphasises built-up materials relative to vegetated or water surfaces and allows regional calibration through the coefficient c. This colour space approach is scalable to global datasets such as MODIS and Sentinel-2 and performs well in vegetated or mixed land cover mosaics[26]. Limitations stem from the HSV transform and from radiometric effects. Hue is unstable at low saturation and can wrap around, while value is sensitive to illumination and shadows. Results also depend on the choice of c, sensor bit depth and radiometric normalisation. Confusion may occur over bright dry soils, salt or sand or metallic and painted roofs, and SWIR availability is required. Per scene tuning of c, light radiometric normalisation, water and vegetation masks and a small ε in the denominator when needed can improve robustness.

$$\mathrm{RGB}_{\mathrm{composite}} = (\rho_{\mathrm{NIR}}, \rho_{\mathrm{SWIR}}, c \times \rho_{\mathrm{Red}}) \tag{45}$$

$$\mathrm{UBI} = \frac{H-V}{H+V} \tag{46}$$

*26) **MNDSI (Modified Normalized Difference Soil Indice)**:* MNDSI[27] enhances soil and urban separation by contrasting SWIR_2 reflectance with the high resolution panchromatic (PAN) band in Landsat 8. The ratio amplifies pixels where soil shows relatively stronger SWIR_2 than broadband PAN and improves discrimination in peri urban mosaics. Reported separability is strong with SDI equal to 2.842 in test areas[27]. Strengths include good soil and impervious contrast and usefulness as a preprocessing mask. Limitations arise from sensor specificity, since the method requires a PAN band and is not available on Sentinel-2, from sensitivity of PAN to illumination, shadows and haze and from the spatial resolution mismatch between SWIR_2 at about 30 m and PAN at 15 m. This mismatch requires resampling and can introduce artefacts. Confusion may also persist over bright sands, salt flats or highly reflective roofs. Radiometric normalisation, shadow and water masks, careful resampling to the SWIR grid and data driven thresholding such as Otsu or k-means can improve robustness.

$$\mathrm{MNDSI} = \frac{\rho_{\mathrm{SWIR}_2} - \rho_{\mathrm{PAN}}}{\rho_{\mathrm{SWIR}_2} + \rho_{\mathrm{PAN}}} \tag{47}$$

*27) **RUI (Ratio Urban Indice)**:* RUI[27] reduces soil confusion by dividing a built up enhancer (BCI) by a soil sensitive term (MNDSI) after min to max normalisation of MNDSI. Pixels with strong urban signal, that is high BCI, but weak soil response, that is low $\mathrm{MNDSI}_{\mathrm{norm}}$, are amplified, while bright soils are attenuated. Reported separability is strong with SDI equal to 1.766 for urban versus soil and 2.542 for urban versus vegetation and TD equal to 1587/1925[27]. Strengths include improved discrimination in semi arid settings and compatibility with Landsat 8. Limitations stem from dependencies on MNDSI inputs, which require SWIR_2 and a PAN band and where resampling can introduce artefacts, on BCI inputs through TCT coefficients and normalisation and on denominator stability when $\mathrm{MNDSI}_{\mathrm{norm}} \approx 0$. Sensitivity to haze and shadows and to scene wise min to max scaling can also affect transferability. In practice, it is helpful to guard the denominator with a small ε, use robust percentile based normalisation for MNDSI, apply water and vegetation masks and adopt data driven thresholds, for example Otsu or k-means, if a binary map is required.

$$\mathrm{RUI} = \frac{\mathrm{BCI}}{\mathrm{MNDSI}_{\mathrm{norm}}} \tag{48}$$

*28) **NRUI (Normalized Ratio Urban Indice)**:* NRUI[27] further suppresses soil influence by normalising the contrast between a built up enhancer (RUI) and the soil sensitive term ($\mathrm{MNDSI}_{\mathrm{norm}}$). Pixels with high RUI and low $\mathrm{MNDSI}_{\mathrm{norm}}$ are amplified, while bright soils are attenuated. Among evaluated indices, NRUI reported the highest urban versus soil separability with SDI equal to 2.224 and TD equal to 1903, while maintaining strong performance in arid and semi urban regions. Urban versus vegetation separability was slightly lower with SDI equal to 1.593[27]. Strengths include robust soil suppression and compatibility with Landsat 8. Limitations arise from dependencies on MNDSI inputs, which require SWIR_2 and a PAN band and can suffer from resolution mismatch and resampling artefacts, on RUI components that depend on BCI and sensor specific TCT coefficients and normalisation and on denominator stability when $\mathrm{RUI} + \mathrm{MNDSI}_{\mathrm{norm}} \approx 0$. Sensitivity to haze and shadows and outliers in scene wise min to max scaling can also affect transferability. Guarding the denominator with a small ε, using robust percentile normalisation for MNDSI, applying water and vegetation masks and adopting data driven thresholds, for example Otsu or k-means, are recommended.

$$\mathrm{NRUI} = \frac{\mathrm{RUI} - \mathrm{MNDSI}_{\mathrm{norm}}}{\mathrm{RUI} + \mathrm{MNDSI}_{\mathrm{norm}}} \tag{49}$$

*29) **NDISI (Normalized Difference Impervious Surface Index)**:*

*a) **$\mathrm{NDISI}_{\mathrm{VIS}}$ (using visible band)**:* $\mathrm{NDISI}_{\mathrm{VIS}}$[28] contrasts thermal emission with the mean of visible, NIR and SWIR_1 reflectance in a normalised difference. It uses the fact that built up surfaces are often thermally warmer than surrounding vegetation and water, while avoiding explicit vegetation masking. Applied to Landsat ETM+ over Fuzhou, it achieved 90.7% overall accuracy with a Kappa of 0.812 and showed strong agreement with impervious area estimates with $R^2 = 0.9054$ and RMSE equal to 0.0787[28]. Strengths include a direct thermal and spectral contrast and simple computation. Limitations include dependence on a thermal band, which is not available on Sentinel 2, coarse TIR spatial resolution and resampling and registration to the reflective bands, diurnal and seasonal variability in surface temperature and atmospheric effects such as humidity and haze and confusion over water, wetlands or cool, reflective roofs. In practice, using a water mask, for example MNDWI, aligning all bands to the TIR grid and applying data driven thresholds such as Otsu or k-means improves robustness.

$$\mathrm{NDISI}_{\mathrm{VIS}} = \frac{\rho_{\mathrm{TIR}} - (\rho_{\mathrm{VIS}_1} + \rho_{\mathrm{NIR}} + \rho_{\mathrm{SWIR}_1})/3}{\rho_{\mathrm{TIR}} + (\rho_{\mathrm{VIS}_1} + \rho_{\mathrm{NIR}} + \rho_{\mathrm{SWIR}_1})/3} \tag{50}$$

b) $NDISI_{WI}$ (using water index): $\mathrm{NDISI_{WI}}$[28] adapts NDISI by replacing the visible band term with a water indice (WI, for example NDWI or MNDWI). It forms a normalised contrast between thermal emission and the mean of WI, NIR and $\mathrm{SWIR_1}$. This substitution reduces water related false positives in riverine, deltaic and coastal settings and removes the need for an explicit water mask. Applied to ASTER imagery over Xiamen, $\mathrm{NDISI_{WI}}$ achieved 90.8% overall accuracy with a Kappa of 0.815 and showed strong agreement with impervious area estimates with $R^2 = 0.9054$ and RMSE equal to 0.0787[28]. Strengths include direct thermal and spectral discrimination and better handling of water adjacent urban fabric. Limitations include dependence on a thermal band with coarser resolution and calibration, sensitivity to the choice and parameterisation of the water indice, for example NDWI versus MNDWI and turbid or shallow waters, diurnal and seasonal temperature variability and registration and resampling between TIR and reflective bands. In practice, align all inputs to the TIR grid, choose WI to match local water conditions, often MNDWI in built up areas, and use data driven thresholds such as Otsu or k-means for robust binarisation.

$$\mathrm{NDISI_{WI}} = \frac{\rho_{\mathrm{TIR}} - (\mathrm{WI} + \rho_{\mathrm{NIR}} + \rho_{\mathrm{SWIR_1}})/3}{\rho_{\mathrm{TIR}} + (\mathrm{WI} + \rho_{\mathrm{NIR}} + \rho_{\mathrm{SWIR_1}})/3} \tag{51}$$

30) EBBI (Enhanced Built-Up and Bareness Indice): EBBI[29] separates built up from bare surfaces by combining a $\mathrm{SWIR_1}$ and NIR contrast with a thermal term in a normalised form. In this way it uses both spectral and temperature differences without explicit vegetation or water masks. In Denpasar, with IKONOS reference, EBBI reported 66.24% overall accuracy, with built up and barren accuracies of 69.65% and 75.11% and a correlation with built up area of $r = 0.70$ with RMSE equal to 49.71% and MBE equal to 25.37%[29]. Strengths include simplicity, use of widely available bands on thermal capable sensors and improved discrimination over purely spectral indices such as NDBI and IBI in some settings. Limitations include dependence on a thermal band that often has coarser resolution than reflective bands, sensitivity to emissivity and atmospheric conditions such as humidity and haze, inter sensor calibration and possible confusion with warm bright soils or industrial roofs. Registration and resampling between TIR and SWIR or NIR can also affect performance. Aligning reflective bands to the TIR grid, applying basic radiometric and atmospheric normalisation and using data driven thresholds such as Otsu or k-means when a binary mask is required can improve robustness.

$$\mathrm{EBBI} = \frac{\rho_{\mathrm{SWIR_1}} - \rho_{\mathrm{NIR}}}{10 \times \sqrt{\rho_{\mathrm{SWIR_1}} + \rho_{\mathrm{TIR}}}} \tag{52}$$

31) MNDISI (Modified Normalized Difference Impervious Surface Indice): MNDISI[30] augments impervious surface detection by contrasting emissivity corrected land surface temperature (T_s) with a mean spectral term that includes MNDWI, ρ_{NIR} and $\rho_{\mathrm{SWIR_1}}$ in a normalised difference. The thermal component directly captures heat retention in built up fabric, while the spectral part suppresses water and vegetation responses. Emissivity is estimated from NDVI to refine T_s. Reported results show improved accuracy over NDISI in complex urban scenes[30]. Strengths include reduced reliance on explicit vegetation masks and better robustness to spectral ambiguity. Limitations arise from dependence on a thermal band with coarser resolution than reflective bands and on accurate emissivity and LST retrieval, sensitivity to diurnal and seasonal temperature variability and atmospheric conditions such as humidity and haze, co registration and resampling between TIR and reflective bands and the quality of NDVI and MNDWI under shadows or turbidity. Align bands to the TIR grid, apply basic atmospheric and radiometric normalisation, choose a water indice suited to local waters, for example NDWI or MNDWI, and use data driven thresholds such as Otsu or k-means for binarisation.

$$\mathrm{MNDISI} = \frac{T_s - (\mathrm{MNDWI} + \rho_{\mathrm{NIR}} + \rho_{\mathrm{SWIR_1}})/3}{T_s + (\mathrm{MNDWI} + \rho_{\mathrm{NIR}} + \rho_{\mathrm{SWIR_1}})/3} \tag{53}$$

$$T_s = \frac{T_b}{1 + \left(\frac{\lambda T_b}{\rho}\right) \ln \varepsilon}, \tag{54}$$

where $\lambda = 10.8\,\mu\mathrm{m}$, $\rho = 1.438 \times 10^{-2}\,\mathrm{m \cdot K}$ and ε is emissivity derived from NDVI:

$$\varepsilon = \begin{cases} 0.979 - 0.035 \cdot \rho_{\mathrm{Red}}, & \mathrm{NDVI} < 0.2, \\ 0.986 + 0.004 \cdot P_v, & 0.2 \le \mathrm{NDVI} \le 0.5, \\ 0.990, & \mathrm{NDVI} > 0.5, \end{cases} \tag{55}$$

$$P_v = \left[\frac{\mathrm{NDVI} - 0.2}{0.5 - 0.2}\right]^2. \tag{56}$$

32) STRed (SWIR–TIR–Red Indice): STRed[16] combines a shortwave and red reflective term with a thermal term in a normalised contrast to separate classes in spectrally complex transitional environments, for example Mediterranean and semi arid landscapes. By integrating reflectance and surface temperature cues, it helps distinguish vegetation, water, bare soils and mining zones and complements the NIR free SwiRed within the LICA scheme[16]. In practice, atmospherically corrected Landsat 5, 7 or 8 imagery is used, bands are min to max normalised and empirical thresholds are applied for mapping. Strengths include use of thermal contrast where purely spectral indices struggle. Limitations include reliance on a thermal band that has coarser resolution than reflective bands and is not available on Sentinel 2, sensitivity to diurnal and seasonal LST variability and emissivity assumptions, atmospheric effects such as humidity and haze and registration and resampling between TIR and SWIR or Red. STRed is also less targeted to direct built up detection than dedicated urban indices. Careful co registration, scene wise normalisation, water and vegetation masks and data driven thresholds such as Otsu or k-means can improve robustness.

$$\mathrm{STRed} = \frac{\rho_{\mathrm{SWIR_1}} + \rho_{\mathrm{Red}} - \rho_{\mathrm{TIR_1}}}{\rho_{\mathrm{SWIR_1}} + \rho_{\mathrm{Red}} + \rho_{\mathrm{TIR_1}}} \tag{57}$$

33) $\mathrm{NDBI_{OLI}}$ (Normalized Difference Built-up Indice for OLI): $\mathrm{NDBI_{OLI}}$[31] augments the classical NDBI by incorporating principal components from SWIR bands 6 and 7 and thermal bands 10 and 11 alongside NIR band 5. After radiometric and atmospheric correction of the reflective bands and preparation of LST or brightness temperature for TIR, per scene PCAs are computed over $\{B6, B7\}$

and $\{B10, B11\}$. The leading components, $PCA_{6,7}$ and $PCA_{10,11}$, capture material and temperature contrasts that, combined with $B5$, sharpen impervious surface discrimination:

$$\mathrm{NDBI_{OLI}} = \frac{(PCA_{6,7} + PCA_{10,11}) - \rho_{B5}}{(PCA_{6,7} + PCA_{10,11}) + \rho_{B5}} \tag{58}$$

Strengths include improved separation in thermally heterogeneous scenes and better suppression of vegetation and soil confusion compared with purely spectral NDBI. Limitations arise from dependence on a thermal band with coarser native resolution where resampling and registration to 30 m can introduce artefacts, scene dependent PCA loadings that reduce interpretability and cross scene comparability, sensitivity to emissivity and LST estimation and atmospheric conditions and sensor specificity to Landsat 8 and 9 OLI and TIRS. In practice, align all inputs to a common grid, often the TIR grid, use robust scaling before PCA, document PCA loadings for reproducibility and employ data driven thresholds such as Otsu or k-means if a binary map is required.

*34) **BAEM (Built-up Area Extraction Method):*** BAEM[31] integrates a PCA based built up enhancer ($\mathrm{NDBI_{OLI}}$) with vegetation and water suppressors (NDVI, MNDWI) to reduce spectral confusion in mixed urban, barren and vegetation scenes. The workflow applies radiometric and atmospheric correction and optional 15 m sharpening, computes per scene PCAs on SWIR (B6–B7) and TIR (B10–B11) to form $\mathrm{NDBI_{OLI}}$, derives $\mathrm{NDVI_{OLI}}$ and $\mathrm{MNDWI_{OLI}}$, and then segments the composite with Density Function Peak Search (DFPS) for adaptive thresholding. This combination exploits thermal and spectral contrasts while directly suppressing vegetation and water responses. Strengths include improved discrimination in thermally heterogeneous urban fabrics and reduced manual tuning. Limitations include dependence on a thermal band with coarser native resolution, where resampling and registration can introduce artefacts, scene dependent PCA loadings that reduce interpretability and transferability, sensitivity to emissivity or LST estimation and atmospheric conditions, residual confusion over bright bare soils and the choice of DFPS parameters. In practice, align all inputs to a common grid, often the TIR grid, use robust scaling before PCA, mask clouds, shadows and water, and record PCA loadings and thresholds for reproducibility.

$$\mathrm{BAEM_{OLI}} = \mathrm{NDBI_{OLI}} - \mathrm{NDVI_{OLI}} - \mathrm{MNDWI_{OLI}}, \tag{59}$$

$$\mathrm{NDVI_{OLI}} = \frac{\rho_{B5} - \rho_{B4}}{\rho_{B5} + \rho_{B4}}, \quad \mathrm{MNDWI_{OLI}} = \frac{\rho_{B3} - \rho_{B7}}{\rho_{B3} + \rho_{B7}} \tag{60}$$

*35) **NDII (Normalized Difference Impervious Indice):*** NDII[32] contrasts a visible band reflectance, typically Red, with thermal emission to highlight impervious surfaces and reduce confusion with vegetation and shadows. In practice, Landsat TM reflective bands are radiometrically corrected, the thermal channel is converted to brightness temperature and co registered to the reflective grid, and NDII is thresholded with a scene adaptive histogram method[32]. Strengths include simplicity and direct use of thermal and spectral contrast. Limitations include reliance on a thermal band, which has coarser native resolution and is sensitive to emissivity and atmospheric effects, the need to scale visible and thermal terms to comparable ranges before ratioing, possible registration and resampling artefacts, diurnal and seasonal LST variability and residual confusion with bright bare soils or reflective roofs. Water masking, for example with MNDWI, light radiometric normalisation and data driven thresholds such as Otsu or k-means can improve robustness.

$$\mathrm{NDII} = \frac{\rho_{\mathrm{Vis}} - \rho_{\mathrm{TIR}}}{\rho_{\mathrm{Vis}} + \rho_{\mathrm{TIR}}} \tag{61}$$

*36) **DBI (Dry Built-up Indice):*** DBI[33] targets arid zone urban fabric by contrasting a visible Blue reflectance, often high over impervious low moisture materials, with thermal emission and then subtracting NDVI to suppress vegetation. This yields strong responses over dry built surfaces. In practice, Landsat 8 dry season imagery is radiometrically corrected, the thermal channel is converted to brightness temperature and co registered to the reflective grid, DBI is computed and a scene specific threshold is applied, for example $\tau \approx 0.72$ in dry season tests[33]. Strengths include good performance in arid environments and explicit vegetation suppression. Limitations include Blue band sensitivity to haze and illumination, dependence on thermal resolution and emissivity and atmospheric conditions, diurnal and seasonal LST variability, possible over correction where vegetation is mixed with built up areas such as green roofs or street trees and residual confusion with bright sands, salt flats or reflective roofs. Water masking with MNDWI, careful band co registration, light radiometric normalisation and data driven thresholds such as Otsu or k-means can improve robustness.

$$\mathrm{DBI} = \frac{\rho_{\mathrm{Blue}} - \rho_{\mathrm{TIR_1}}}{\rho_{\mathrm{Blue}} + \rho_{\mathrm{TIR_1}}} - \mathrm{NDVI} \tag{62}$$

*37) **EUBI (Enhanced Urban Built-up Indice):*** EUBI[26] fuses nighttime lights with optical reflectance by taking the ratio of a bright tail VIIRS DNB radiance composite, for example annual or 90th percentile NTL, to the annual maximum UBI derived from HSV transformed MODIS MCD43A4 reflectance. The fusion amplifies persistently lit, spectrally built surfaces and reduces overglow and saturation issues that appear in single source indices. In practice, MODIS reflectance is BRDF and atmosphere corrected, VIIRS radiance is composited to reduce transient artefacts, snow and gas flares are masked with NDSI or ancillary layers and grid based thresholds are applied for mapping[26]. Strengths include improved detection in under lit or spectrally complex urban fabric and robustness over large areas. Limitations arise from multi sensor fusion, including spatial resolution mismatch between VIIRS, about 375 m, and MODIS, about 500 m, co registration and temporal alignment, regional lighting variability such as curfews, outages and cultural lighting, blooming and stray light from bright non residential emitters such as industrial sites and flares, cloud contamination in MODIS and sensitivity to the chosen NTL percentile and annual maxima. Practical mitigations include aligning to a common grid based on the coarser sensor, winsorising or top coding extreme NTL, explicit flare and bright water masking, use of a robust NTL percentile rather than a hard maximum and data driven thresholds. A small ε can be added to the denominator if numerical stability is needed.

$$\mathrm{EUBI} = \frac{\mathrm{NTL}_{\max}}{\mathrm{UBI}_{\max}} \tag{63}$$

38) ***HSI (Human Settlement Indice):*** HSI[34] estimates fractional settlement intensity by combining a normalised DMSP OLS nighttime lights radiance, $\mathrm{OLS}_{\mathrm{nor}}$, with peak vegetation greenness, $\mathrm{NDVI}_{\max}$. Bright, persistently lit pixels raise the indice, while abundant vegetation lowers it. In practice, $\mathrm{NDVI}_{\max}$ is derived with maximum value compositing from multi date MODIS NDVI, OLS radiance is scaled to $[0, 1]$ and the HSI formula below yields a continuous surface. Validations against high resolution imagery report improved mapping in vegetated and dimly lit areas[34]. Strengths include simplicity, global coverage and mitigation of overglow and omission errors compared with NTL only approaches. Limitations stem from the coarse, saturated and blooming prone OLS sensor, including spatial smoothing and inter annual calibration, from sensitivity of $\mathrm{NDVI}_{\max}$ to seasonality and clouds and from reliance on per scene normalisation. Bright non residential emitters such as industrial sites and flares and very dark but built areas can still bias estimates. Practical mitigations include using higher resolution NTL where available, for example VIIRS DNB, applying flare and water masks, constraining NDVI compositing to appropriate seasons and using robust percentile based normalisation. A small ε can be added to denominators if numerical stability is needed.

$$\mathrm{OLS}_{\mathrm{nor}} = \frac{\mathrm{OLS} - \mathrm{OLS}_{\min}}{\mathrm{OLS}_{\max} - \mathrm{OLS}_{\min}}, \tag{64}$$

$$\mathrm{NDVI}_{\max} = \max\left(\mathrm{NDVI}_1, \mathrm{NDVI}_2, \ldots, \mathrm{NDVI}_n\right), \tag{65}$$

$$\mathrm{HSI} = \frac{(1 - \mathrm{NDVI}_{\max}) + \mathrm{OLS}_{\mathrm{nor}}}{(1 - \mathrm{OLS}_{\mathrm{nor}}) + \mathrm{NDVI}_{\max} + \mathrm{OLS}_{\mathrm{nor}} \cdot \mathrm{NDVI}_{\max}} \tag{66}$$

39) ***VANUI (Vegetation Adjusted NTL Urban Indice):*** VANUI[35] highlights brightly lit, low vegetation urban fabric by multiplying vegetation scarcity, given by $1-\mathrm{NDVI}$, with normalised nighttime lights. In practice, NDVI is derived from MODIS NBAR reflectance and negative NDVI from water or shadow is often clipped to zero. DMSP or OLS radiance is scaled to $[0, 1]$ before combination. This simple fusion improves delineation of dense urban cores compared with NTL only products. Limitations include the coarse OLS footprint and saturation and blooming that cause overglow, temporal and geometric mismatch between NTL and NDVI, sensitivity of NDVI to seasonality and clouds and weaker performance in vegetated suburbs or arid cities where NDVI and urban patterns are weakly related. Preprocessing steps such as flare and bright water masking, robust NTL scaling with percentiles, top coding or log transforms, seasonal NDVI compositing and careful co registration improve robustness. VIIRS DNB can replace OLS where available.

$$\mathrm{VANUI} = (1 - \mathrm{NDVI}) \times \mathrm{NTL} \tag{67}$$

40) ***LISI (Large-scale Impervious Surface Indice):*** LISI[36] estimates impervious coverage by multiplying vegetation scarcity with stabilised nighttime light intensity. The term $1 - \mathrm{NDVI}_{\max}$ from seasonal MODIS composites is fused with the square root of min to max normalised VIIRS DNB radiance to reduce saturation and blooming in brightly lit cores while retaining contrast in peri urban and mixed use areas. The square root transform compresses the bright tail of NTL, improves dynamic range and mitigates overglow relative to a linear scale[36]. Strengths include global coverage and improved detection where either NDVI or NTL alone is insufficient. Limitations include spatial and temporal mismatch between MODIS, about 500 m, and VIIRS, about 375 m, sensitivity to lighting variability such as curfews, outages and cultural lighting, residual contamination from bright non residential emitters such as industrial sites and flares and underestimation over dimly lit but impervious surfaces such as new developments or low income housing. Robust preprocessing that includes seasonal NDVI compositing, flare and bright water masking, percentile based NTL normalisation, careful co registration to a common grid and optional data driven thresholds can improve transferability.

$$\mathrm{LISI} = (1 - \mathrm{NDVI}_{\max}) \times \sqrt{\mathrm{DNB}_{\mathrm{norm}}}, \tag{68}$$

where

$$\mathrm{DNB}_{\mathrm{norm}} = \frac{\mathrm{DNB} - \mathrm{DNB}_{\min}}{\mathrm{DNB}_{\max} - \mathrm{DNB}_{\min}}. \tag{69}$$

41) ***NDUI (Normalized Difference Urban Indice):*** NDUI[37] fuses nighttime illumination with vegetation scarcity by forming a normalised difference between normalised nightlight intensity and seasonal NDVI, clipped to non negative values. The index yields higher values where lighting is strong and vegetation is sparse, which is typical of impervious urban cores. In practice, DMSP or OLS, or VIIRS DNB where available, is radiometrically normalised and co registered to the NDVI grid. NDVI is composited seasonally to reduce cloud and phenology noise, clipped at $\mathrm{NDVI} \geq 0$ and NDUI is computed and thresholded for mapping[37]. Strengths include simplicity, improved boundary detail compared with NTL only or NDVI only products and scalability to coarse and medium resolution imagery. Limitations arise from the coarse and blooming prone OLS sensor with spatial smoothing and saturation, temporal and geometric misalignment between NTL and NDVI, weaker performance in vegetated suburbs or dimly lit impervious areas and instability when $\mathrm{NTL} + \mathrm{NDVI}$ is very small. Bright non residential emitters such as industrial sites and flares can also bias results. Recommended mitigations include preferring VIIRS DNB when possible, applying flare and bright water masks, using seasonal NDVI that matches local phenology, co registering to a common grid, top coding or percentile normalising NTL and adding a small ε to the denominator if needed.

$$\mathrm{NDUI} = \frac{\mathrm{NTL} - \mathrm{NDVI}}{\mathrm{NTL} + \mathrm{NDVI}}, \quad \mathrm{NDVI} \geq 0 \tag{70}$$

42) ***NUACI (Normalized Urban Areas Composite Indice):*** NUACI[38] reduces DMSP OLS overglow and bright non urban false positives by gating nightlight intensity in a biophysical space. Pixels are kept only if their MODIS derived $(\mathrm{NDWI}, \mathrm{EVI}_{\max})$ lie within a radius r of an urban centroid (a, b). Surviving pixels are then scaled by their spectral proximity and the min to max normalised OLS radiance. Operationally, seasonal or annual $\mathrm{EVI}_{\max}$ and NDWI are composited, the urban centroid and radius are estimated from training data or clustering and NUACI is computed with the distance d in (72) and the rule in (71). Strengths include explicit suppression of bright non urban emitters such as roads, water and flares and improved urban delineation in raw OLS data. Limitations include dependence on reliable training data and the choice of r, which control transferability, coarse native resolutions for MODIS and OLS, temporal and geometric misalignment between NTL and MODIS and reduced sensitivity to dimly lit impervious areas. Performance also varies with seasonal vegetation and water dynamics. Practical mitigations include preferring VIIRS DNB where available, constraining compositing windows to local phenology, applying flare and bright water masks, co registering to a common grid and validating (a, b, r) with held out samples.

$$\mathrm{NUACI} = \begin{cases} 0, & d > r, \\ \left(1 - \frac{d}{r}\right) \times \frac{\mathrm{OLS} - \mathrm{OLS}_{\min}}{\mathrm{OLS}_{\max} - \mathrm{OLS}_{\min}}, & d \le r, \end{cases} \tag{71}$$

$$d = \sqrt{(\mathrm{NDWI} - a)^2 + (\mathrm{EVI}_{\max} - b)^2} \tag{72}$$

43) ***NAISI (Nighttime Lights Adjusted Impervious Surface Indice):*** NAISI[39] fuses illumination and moisture or structure cues while suppressing vegetation. It averages a normalised nighttime lights component, $\text{NTL-PC1}_{\mathrm{nor}}$, which is the first principal component from high resolution ISS NTL composites, with normalised Tasseled Cap wetness, $\mathrm{TC3}_{\mathrm{nor}}$, and subtracts normalised SAVI. Brightly lit, relatively dry built pixels yield high $\text{NTL-PC1}_{\mathrm{nor}}$ and low SAVI and hence higher NAISI near urban cores. Vegetated or water dominated pixels are down weighted by the SAVI and TC3 terms. Operationally, optical bands are atmospherically corrected, TC3 is computed with sensor specific TCT coefficients and min to max normalised, ISS NTL scenes are radiometrically harmonised and reduced with PCA to PC1 then normalised and SAVI is formed and scaled before combination[39]. Strengths include reduced overglow compared with raw NTL and improved discrimination along urban and rural boundaries. Limitations stem from multi sensor fusion, including spatial and temporal misalignment between NTL and optical data, scene dependent PCA loadings that affect interpretability and transferability, outlier sensitivity in min to max normalisation, possible sign and convention differences in TC3 across implementations and residual blooming from bright non residential emitters such as industrial sites and flares. Dimly lit impervious areas may be underestimated. Mitigations include co registering to a common grid based on the coarser sensor, percentile based scaling, flare and bright water masks, documentation of PCA loadings and, where ISS products are unavailable, substitution of VIIRS DNB composites.

$$\mathrm{NAISI} = \frac{\text{NTL-PC1}_{\mathrm{nor}} + \mathrm{TC3}_{\mathrm{nor}}}{2} - \mathrm{SAVI}_{\mathrm{nor}}, \tag{73}$$

$$\begin{aligned} \mathrm{TC3} = {} & 0.1511\rho_B + 0.1973\rho_G + 0.3283\rho_R + 0.3407\rho_{\mathrm{NIR}} \\ & - 0.7117\rho_{\mathrm{SWIR}_1} - 0.4559\rho_{\mathrm{SWIR}_2}. \end{aligned} \tag{74}$$

44) $\mathrm{MNDISI}_{\mathrm{Liu}}$ ***(Modified Normalized Difference Impervious Surface Indice, Liu):*** $\mathrm{MNDISI}_{\mathrm{Liu}}$[30] fuses emissive, reflective and illumination cues by contrasting the sum of land surface temperature, T_{LST}, and a normalised nighttime lights term, L_{LIT}, against a water and spectral suppressor, $\mathrm{MNDWI} + \rho_{\mathrm{SWIR}_1}$, in a normalised difference. Thermally warm and consistently lit impervious pixels raise the numerator, while water and vegetation responses lower the denominator. This improves urban detection where purely spectral indices struggle. In applications over Chinese megacities such as Shenzhen and Guangzhou it achieved about 87% overall accuracy with Kappa around 0.74[30]. Strengths include complementary sensing that combines TIR, NTL and optical data and robustness in spectrally complex fabric. Limitations arise from reliance on a thermal band with coarser native resolution and sensitivity to emissivity and atmospheric conditions, NTL calibration and blooming or overglow from industrial emitters and flares, co registration and scale mismatch among TIR, NTL and reflective grids and sensitivity to how L_{LIT} and MNDWI are normalised, including outliers and seasonality. Practical mitigations include aligning all inputs to a common grid, often the TIR grid, preferring VIIRS DNB for L_{LIT} when available, masking clouds, shadows and water, using percentile based normalisation and applying data driven thresholds such as Otsu or k-means for binarisation.

$$\mathrm{MNDISI}_{\mathrm{Liu}} = \frac{T_{\mathrm{LST}} + L_{\mathrm{LIT}} - (\mathrm{MNDWI} + \rho_{\mathrm{SWIR}_1})}{T_{\mathrm{LST}} + L_{\mathrm{LIT}} + (\mathrm{MNDWI} + \rho_{\mathrm{SWIR}_1})} \tag{75}$$

45) ***VTLI (Vegetation–Temperature–Light Indice):*** VTLI[40] fuses vegetation scarcity, land surface temperature and nighttime illumination to highlight impervious urban fabric and reduce saturation and blooming effects in legacy DMSP OLS products. Operationally, annual maximum NDVI from MODIS is inverted to $1 - \mathrm{NDVI}_{\max}$, maximum monthly daytime LST is scaled to $T_{\max} \in [0, 1]$ and an annual bright tail NTL composite is normalised to $L \in [0, 1]$. Their product yields a graded surface that strengthens persistent, warm, well lit urban cores and captures urban and rural gradients[40]. Reported validations across Chinese cities showed improvements over NTL only methods, with κ values of 0.410 in Beijing, 0.718 in Shanghai, 0.483 in Lanzhou and 0.623 in Shenyang[40]. Strengths include simplicity, complementary sensing that combines optical, thermal and NTL data and large area scalability. Limitations stem from multi sensor fusion, including spatial and temporal misalignment between MODIS and OLS, the coarse and blooming prone OLS footprint, diurnal and seasonal LST variability and emissivity effects and sensitivity to the use of maxima that can introduce outliers and cloud artefacts. Dimly lit impervious areas and vegetated built up zones can be underestimated. Mitigations include substituting VIIRS DNB for OLS where available, using seasonal NDVI and LST compositing matched to local phenology, applying flare and bright water masking, robust percentile based normalisation, co registration to a common grid and, if needed, data driven thresholds for binarisation.

$$\mathrm{VTLI} = (1 - \mathrm{NDVI}_{\max}) \times T_{\max} \times L \tag{76}$$

46) ***TVANUI (Temperature and Vegetation Adjusted NTL Urban Indice):*** TVANUI[41] combines thermal, vegetation and illumination cues by mapping the ratio $\mathrm{LST}/(\mathrm{NDVI} + \varepsilon)$ through an arctangent and multiplying the result by normalised nighttime lights. The arctangent stabilises extreme values in vegetated or bare areas. Using MODIS annual maxima for NDVI and LST and scaled DMSP OLS NTL, TVANUI produced strong results in global case studies. For example, Beijing showed overall accuracy of 94.05% with $\kappa = 0.681$, New York City showed $\kappa = 0.511$ and the correlation with mapped urban extent was $R = 0.831$[41]. Strengths include robustness to vegetation and soil variability and improved urban delineation compared with NTL only indices. Limitations arise from the coarse and saturated OLS sensor with blooming, temporal and geometric mismatch among NDVI, LST and NTL, diurnal and seasonal LST variability and emissivity assumptions, sensitivity to the choice of ε and to hot non urban surfaces such as bare rock or industrial grounds and under detection in dimly lit impervious areas. Mitigations include preferring VIIRS DNB instead of OLS where possible, co registering all inputs to a common grid, selecting a small ε, for example between 10^{-3} and 10^{-2}, to guard division, using seasonal compositing matched to local phenology, applying flare and bright water masks and adopting data driven thresholds such as Otsu or k-means for binarisation.

$$\text{TVANUI} = \left(\frac{\arctan\left(\frac{\text{LST}}{\text{NDVI}+\epsilon}\right)}{\pi/2} \right) \times \left(\frac{\text{NTL}}{63} \right) \quad (77)$$

REFERENCES

[1] Y. Zha, J. Gao, and S. Ni, "Use of normalized difference built-up index in automatically mapping urban areas from tm imagery," *International Journal of Remote Sensing*, vol. 24, no. 3, pp. 583–594, 2003.

[2] H. Xu, "A new index for delineating built-up land features in satellite imagery," *International Journal of Remote Sensing*, vol. 29, no. 14, pp. 4269–4276, 2008.

[3] R. C. Estoque and Y. Murayama, "Classification and change detection of built-up lands from landsat-7 etm+ and landsat-8 oli/tirs imageries: A comparative assessment of various spectral indices," *Ecological Indicators*, vol. 56, pp. 205–217, 2015.

[4] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960. [Online]. Available: https://doi.org/10.1177/001316446002000104

[5] A. R. et al., "Built-up and bare-land index (bbi) for enhancing urban areas extraction from landsat imagery," *Remote Sensing*, vol. 10, p. 431, 2018.

[6] C. Deng and C. Wu, "Cbci: A combinational biophysical composition index for characterizing urban environments," *Remote Sensing of Environment*, vol. 221, pp. 108–121, 2019.

[7] S. Li, L. Zhang, H. Shen, Q. Liu, P. He, and Y. Zhang, "An overview of the gaofen series satellites and their applications for earth observation," *Remote Sensing*, vol. 10, no. 6, pp. 1–22, 2018. [Online]. Available: https://doi.org/10.3390/rs10060872

[8] Y. Bai, G. He, G. Wang, and G. Yang, "We-ndbi-a new index for mapping urban built?up areas from gf?1 wfv images," *Remote Sensing Letters*, vol. 11, no. 5, pp. 407–415, 2020.

[9] Y. Tian, H. Chen, Q. Song, and K. Zheng, "A novel index for impervious surface area mapping: Development and validation," *Remote Sensing*, vol. 10, no. 10, p. 1521, September 2018.

[10] C. Deng and C. Wu, "Bci: A biophysical composition index for remote sensing of urban environments," *Remote Sensing of Environment*, vol. 127, pp. 247–259, 2012. [Online]. Available: https://doi.org/10.1016/j.rse.2012.09.009

[11] T. Y. Kawamura Makoto, Jayamanna Sanath, "Quantitative assessment of urbanization in developing countries using satellite data," *Transactions of the Japan Society of Civil Engineers*, vol. 1, no. 580, pp. 45–54, 1997.

[12] J. Chen, M. Li, Y. Liu, C. Shen, and W. Hu, "Extract residential areas automatically by new built-up index," in *Proceedings of the 2008 International Conference on Remote Sensing*. Changzhou, China: School of Geographic & Oceanographic Science, Nanjing University, 2008, landsat TM-based study of Changzhou using bands 3, 4, and 5.

[13] M. M. Waqar, J. F. Mirza, R. Mumtaz, and E. Hussain, "Development of new indices for extraction of built-up area & bare soil from landsat data," *Scientific Reports*, vol. 1, no. 136, pp. 1–4, 2012. [Online]. Available: https://doi.org/10.4172/scientificreports.136

[14] S. Bouzekri, A. Lasbet, and A. Lachehab, "A new spectral index for extraction of built-up area using landsat-8 data," *Journal of the Indian Society of Remote Sensing*, vol. 43, pp. 46–50, 07 2015.

[15] D. Kaimaris and P. Patias, "Identification and area measurement of the built-up area with the built-up index (bui)," *International Journal of Advanced Remote Sensing and GIS*, vol. 5, no. 6, pp. 1844–1858, 2016, iSSN 2320-0243. [Online]. Available: http://technical.cloud-journals.com/index.php/IJARSG/article/view/Tech-627

[16] A. Capolupo, C. Monterisi, and E. Tarantino, "Landsat images classification algorithm (lica) to automatically extract land cover information in google earth engine environment," *Remote Sensing*, vol. 12, no. 7, p. 1201, 2020. [Online]. Available: https://doi.org/10.3390/rs12071201

[17] J. Chen, K. Yang, S. Chen, C. Yang, S. Zhang, and L. He, "Enhanced normalized difference index for impervious surface area estimation at the plateau basin scale," *Journal of Applied Remote Sensing*, vol. 13, no. 1, p. 016502, 2019. [Online]. Available: https://doi.org/10.1117/1.JRS.13.016502

[18] C. He, P. Shi, D. Xie, and Y. Zhao, "Improving the normalized difference built-up index to map urban built-up areas using a semiautomatic segmentation approach," *Remote Sensing Letters*, vol. 1, no. 4, pp. 213–221, 2010. [Online]. Available: https://doi.org/10.1080/01431161.2010.481681

[19] T. Shi and H. Xu, "Derivation of tasseled cap transformation coefficients for sentinel-2 msi at-sensor reflectance data," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 1–14, 2020. [Online]. Available: https://doi.org/10.1109/JSTARS.2019.2938388

[20] D. Stathakis, K. Perakis, and I. Savin, "Efficient segmentation of urban areas by the vibi," *International Journal of Remote Sensing*, vol. 33, no. 20, pp. 6361–6377, 2012. [Online]. Available: https://doi.org/10.1080/01431161.2012.687842

[21] E. P. Crist and R. C. Cicone, "A physically-based transformation of thematic mapper data—the tm tasseled cap," *IEEE Transactions on Geoscience and Remote Sensing*, vol. GE-22, no. 3, pp. 256–263, 1984.

[22] Y. Deng, C. Wu, M. Li, and R. Chen, "Rndsi: A ratio normalized difference soil index for remote sensing of urban/suburban environments," *International Journal of Applied Earth Observation and Geoinformation*, vol. 39, pp. 40–48, 2015. [Online]. Available: https://doi.org/10.1016/j.jag.2015.02.010

[23] G. Sun, X. Chen, X. Jia, Y. Yao, and Z. Wang, "Combinational build-up index (cbi) for effective impervious surface mapping in urban areas," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 9, no. 5, pp. 2081–2092, 2016. [Online]. Available: https://doi.org/10.1109/JSTARS.2015.2478914

[24] R. Bouhennache, T. Bouden, A. Taleb-Ahmed, and A. Cheddad, "A new spectral index for the extraction of built-up land features from landsat 8 satellite imagery," *Geocarto International*, vol. 34, no. 14, pp. 1531–1551, 2019. [Online]. Available: https://doi.org/10.1080/10106049.2018.1497094

[25] F. Benkouider, A. Abdellaoui, and L. Hamami, "New and improved built-up index using spot imagery: Application to an arid zone (laghouat and m'sila, algeria)," *Journal of the Indian Society of Remote Sensing*, 2019. [Online]. Available: https://doi.org/10.1007/s12524-018-0895-7

[26] R. C. Sharma, R. Tateishi, K. Hara, S. Gharechelou, and K. Iizuka, "Global mapping of urban built-up areas of year 2014 by combining modis multispectral data with viirs nighttime light data," *International Journal of Digital Earth*, vol. 9, no. 10,

pp. 1004–1020, 2016. [Online]. Available: https://doi.org/10.1080/17538947.2016.1168879

[27] A. K. Piyoosh and S. K. Ghosh, "Development of a modified bare soil and urban index for landsat 8 satellite data," *Geocarto International*, vol. 33, no. 4, pp. 423–442, 2018. [Online]. Available: https://doi.org/10.1080/10106049.2016.1273401

[28] H. Xu, "Analysis of impervious surface and its impact on urban heat environment using the normalized difference impervious surface index (ndisi)," *Photogrammetric Engineering and Remote Sensing*, vol. 76, pp. 557–565, 05 2010.

[29] R. A.-S. A. M. A. I. and D. A. I. "Enhanced built-up and bareness index (EBBI) for mapping built-up and bare land in urban area," *Remote Sensing*, vol. 4, no. 10, pp. 2957–2970, 2012.

[30] Z. Sun, C. Wang, H. Guo, and R. Shang, "A modified normalized difference impervious surface index (mndisi) for automatic urban mapping from landsat imagery," *Remote Sensing*, vol. 9, no. 9, 2017. [Online]. Available: https://www.mdpi.com/2072-4292/9/9/942

[31] S. S. Bhatti and N. K. Tripathi, "Built-up area extraction using landsat 8 oli imagery," *GIScience & Remote Sensing*, vol. 51, no. 4, pp. 445–467, 2014. [Online]. Available: https://doi.org/10.1080/15481603.2014.939539

[32] Z. Wang, C. Gang, X. Li, Y. Chen, and J. Li, "Application of a normalized difference impervious index (ndii) to extract urban impervious surface features based on landsat tm images," *International Journal of Remote Sensing*, vol. 36, no. 4, pp. 1055–1069, 2015. [Online]. Available: https://doi.org/10.1080/01431161.2015.1007250

[33] A. Rasul, H. Balzter, G. R. Faqe Ibrahim, H. M. Hameed, J. Wheeler, B. Adamu, S. Ibrahim, and P. M. Najmaddin, "Applying built-up and bare-soil indices from landsat 8 to cities in dry climates," *Land*, vol. 7, no. 3, p. 81, 2018. [Online]. Available: https://doi.org/10.3390/land7030081

[34] D. Lu, H. Tian, G. Zhou, and H. Ge, "Regional mapping of human settlements in southeastern china with multisensor remotely sensed data," *Remote Sensing of Environment*, vol. 112, no. 9, pp. 3668–3679, 2008. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425708001776

[35] Q. Zhang, C. Schaaf, and K. C. Seto, "The vegetation adjusted ntl urban index: A new approach to reduce saturation and increase variation in nighttime luminosity," *Remote Sensing of Environment*, vol. 129, pp. 32–41, 2013. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425712004105

[36] W. Guo, D. Lu, Y. Wu, and J. Zhang, "Mapping impervious surface distribution with integration of snnp viirs-dnb and modis ndvi data," *Remote Sensing*, vol. 7, no. 9, pp. 12 459–12 477, 2015. [Online]. Available: https://www.mdpi.com/2072-4292/7/9/12459

[37] Q. Zhang, B. Li, D. Thau, and R. Moore, "Building a better urban picture: Combining day and night remote sensing imagery," *Remote Sensing*, vol. 7, no. 9, pp. 11 887–11 913, 2015. [Online]. Available: https://doi.org/10.3390/rs70911887

[38] X. Liu, G. Hu, B. Ai, X. Li, and Q. Shi, "A normalized urban areas composite index (nuaci) based on combination of dmsp-ols and modis for mapping impervious surface area," *Remote Sensing*, vol. 7, no. 12, pp. 17 168–17 189, 2015. [Online]. Available: https://doi.org/10.3390/rs71215863

[39] X. Chen, X. Jia, and M. Pickering, "A nighttime lights adjusted impervious surface index (naisi) with integration of landsat imagery and nighttime lights data from international space station," *International Journal of Applied Earth Observation and Geoinformation*, vol. 83, p. 101889, 2019. [Online]. Available: https://doi.org/10.1016/j.jag.2019.05.022

[40] R. Hao, D. Yu, Y. Sun, Q. Cao, Y. Liu, and Y. Liu, "Integrating multiple source data to enhance variation and weaken the blooming effect of dmsp-ols light," *Remote Sensing*, vol. 7, no. 2, pp. 1422–1440, 2015. [Online]. Available: https://doi.org/10.3390/rs70201422

[41] H. G. et al., "A tvanui index for global urban area extraction using cloud computing and sentinel-2 imagery," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 189, pp. 12–24, 2022.